Omar Mohamed Abdelsalam

AI Engineer

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Professional Summary

AI & LLM Engineer with strong experience in Python, Deep Learning, and transformer-based architectures. Skilled in building and deploying multimodal chatbots and Retrieval-Augmented Generation (RAG) systems using Hugging Face, PyTorch, and TensorFlow. Proven ability to deliver production-ready AI solutions with high reliability, leveraging DevOps and MLOps practices for scalable deployment.

Internships

DevOps Engineer Intern Apr 2024 – Oct 2024

Digital Egypt Pioneers Initiative (DEPI) – Alexandria, Egypt (Part-time, Hybrid)

- Deployed AWS infrastructure (EC2, S3, VPC) using Terraform, containerized applications with Docker, orchestrated with Kubernetes, and implemented CI/CD pipelines with GitHub Actions for scalable ML deployments.

Artificial Intelligence Engineer Intern Sep 2023 – Sep 2023

Telecom Egypt, Alexandria, Egypt (Full-Time, Onsite)

- Developed and pre-processed ML models on real-world telecom datasets, performing feature engineering, evaluation, and EDA to improve model performance.

Professional Experience

Python & AI Instructor Jan 2025 – Present

Black Horse Courses – Alexandria, Egypt (Part-time, Onsite)

- Delivered Python and machine learning training to 50+ students, focusing on OOP, data structures, and real-world model deployment with a 95% satisfaction rate.

Skills

- **Machine Learning:** Supervised & Unsupervised Learning, Feature Engineering, Model Evaluation
- **Deep Learning:** CNNs, RNNs, Transformers, Transfer Learning, Model Optimization
- **Computer Vision:** Image Classification, Object Detection, Image Segmentation, OpenCV, YOLO, ResNet
- **Frameworks & Libraries:** PyTorch, TensorFlow, Keras, Scikit-learn
- **Data Processing & Analysis:** NumPy, Pandas, Data Cleaning, Exploratory Data Analysis (EDA)
- **Model Deployment & Serving:** Model Serialization, RESTful APIs, Dockerized Inference, Performance Optimization
- **DevOps & MLOps:** Docker, Kubernetes, Terraform, GitHub Actions, AWS (EC2, S3, IAM), CI/CD for ML Pipelines
- **Programming & Scripting:** Python, Bash, C++, PowerShell
- **Databases & Caching:** MySQL, PostgreSQL, SQL Server, Redis
- **Operating Systems:** Linux (Ubuntu, RHEL), Windows Server
- **Version Control & Collaboration:** Git (GitHub, GitLab)
- **Mathematics for ML:** Linear Algebra, Probability, Statistics, Optimization
- **Computer Science Fundamentals:** OOP, Algorithms, Data Structures
- **Tools & Platforms:** Jupyter Notebook, Google Colab, Linux Tooling
- **Soft Skills:** Leadership, Mentoring, Collaboration, Adaptability

Projects

Gemini Chatbot – Multimodal Assistant – [Link]

- Developed a multimodal AI assistant using Google Gemini LLM.
- Integrated voice, image, and video inputs for rich user interaction.
- Applied Transformers for text generation, sentiment analysis, and named entity recognition (NER).

LocalGPT – Privacy-Preserving Chatbot – [Link]

- Built a privacy-preserving document chatbot using LLaMA2, Hugging Face Transformers, GPTQ, and GGML.
- Designed and deployed Retrieval-Augmented Generation (RAG) pipelines for enhanced document retrieval.
- Delivered fully offline local inference using Streamlit interface and RESTful APIs, ensuring 100% data privacy.

Education

Master of Engineering in Computer Engineering Jan 2026 – Present
Arab Academy for Science, Technology and Maritime Transport (AAST) – Alexandria, Egypt

Diploma in Computer Software Engineering – Intensive Code Camp (ICC) Mar 2025 – Aug 2025
Information Technology Institute (ITI) – Alexandria, Egypt

Bachelor of Engineering in Computer Engineering Sep 2019 – Jul 2024
Higher Institute of Engineering & Technology (BHI) – Alexandria, Egypt

- **CGPA:** 3.2 / 4.0 (Very Good (+))
- **Graduation Project:** HistoryVerse – AI/AR platform for interactive historical exploration (Grade: A+)

Certificates

- **Red Hat Certified System Administrator (RHCSA),** Red Hat — May 2025 [Verified on Red Hat]
- **HCCDA-Tech Essentials Huawei Cloud Developer,** Huawei — Jul 2025 [Verified]

Languages

- **Arabic:** Native
- **English:** Fluent